Ravindra Chowdary Jonnagadla

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Education

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering – Cloud Computing (CGPA: 8.01/10) — May 2026

Technical Skills

Languages: Python, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, Bash, Shell Scripting
AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Whisper, WGAN-GP, PPO (RL), NLP
Cloud & DevOps: AWS (EC2, S3, Lambda, IAM), Docker, Kubernetes, Terraform, CI/CD, Git, Linux
Backend & Web: FastAPI, Flask, Django, Node.js, Next.js, React, RESTful APIs, Microservices, Redis, Celery
Certifications: AWS Certified Cloud Practitioner – Credly Verified Credential (Badge ID: 3dc45737)

Professional Experience

Waani – Conversational Voice AI Platform — Chennai, India
Lead Engineer & Co-Founder — July 2026 – Present

- Architected an end-to-end conversational voice agent integrating telephony protocols, streaming STT (Sarvam AI), and LLM inference, achieving $<$800ms latency for natural Hindi-English code-switching dialogs.
- Engineered conversational state machines and dynamic intent branching for automated lead qualification and appointment scheduling, configuring automated fallback logic for human operator handoff.
- Implemented duplex circular audio buffer management over WebSockets to support real-time user speech interruption and eliminate audio packet drop under network jitter.

LynkZen – Instagram DM Automation SaaS — Chennai, India
Lead Engineer & Co-Founder — April 2026 – June 2026

- Built and launched an Instagram automation SaaS using Next.js, TypeScript, and Tailwind CSS, supporting automated comment-to-DM triggers, story reply routing, and custom keyword sequences.
- Developed an interactive node-based visual workflow builder allowing creators to configure multi-step conditional messaging funnels and automated response paths without writing code.
- Onboarded **3,000 active beta users for closed private testing**, engineering Redis queue handlers to throttle throughput and ensure 100% compliance with Meta Graph API rate limits.

Technical Projects

AMERS: Multimodal Emotion Recognition System | *PyTorch, Signal Processing* — IEEE ICETEA 2026

- Developed a multimodal emotion recognition pipeline combining raw EEG signals and speech features, implementing Cross-Modal Mutual Attention (CMMA) to capture inter-modal correlations.
- Integrated Emotion-Aware Gating (EAG) with WGAN-GP for synthetic EEG feature augmentation, applying Proximal Policy Optimization (PPO) to dynamically calibrate synthetic-to-real sampling ratios.
- Evaluated classification performance across 4 emotion classes on standard benchmarks, achieving **82.55% accuracy**, **0.83 macro F1-score**, and **0.77 Cohen's kappa**.

SmartCap: Real-Time Subtitle Generation | *Python, Whisper, FFmpeg* — IEEE ICICV 2025

- Built an automated audio-visual transcription tool using OpenAI Whisper, implementing word-level acoustic timestamp alignment to generate synchronized video subtitles (.srt/.vtt).
- Engineered media preprocessing scripts using FFmpeg and Python multithreading to isolate acoustic tracks, normalize dynamic audio levels, and burn formatted subtitles onto video.

Smart QR-Based University Parking System | *Flask, Python, Twilio API* — IEEE CONIT 2025

- Developed a campus parking management application with dynamic, time-expiring cryptographic QR code generation for gate validation and Twilio OTP user authentication.
- Implemented a lightweight Progressive Web App (PWA) with real-time slot occupancy tracking, achieving a **98.7% booking transaction success rate** across evaluated trial simulations.

Publications

- **IEEE ICETEA 2026:** *"Adaptive GAN-Augmented EEG–Speech Emotion Recognition via Deep Reinforcement Learning"* – Int. Conf. on Emerging Technologies in Engineering Applications, Co-Author.
- **IEEE ICICV 2025:** *"SmartCap: Real-Time Synchronized Subtitles for Videos"* – Published in IEEE Xplore (Doc ID: 11085491), Sole/Lead Author.
- **IEEE CONIT 2025:** *"Smart QR-Based University Parking Management System"* – Published in IEEE Xplore (Doc ID: 11166870), Co-Author.